

LifeConnect Sri Lanka - IS13

Actor	Function Id	Function	Function description	Business Rule/ Logic	Integration
Public Users	PU001	View	Visitors can view stories and other statistics	<u>Business Rules:</u> View Donation Eligibility Criteria. View Educational content. View Donor Statistics and organ waitlist. <u>Business Logic:</u> View donation eligibility according to the organ. The system must display the latest approved criteria from the Ministry of Health or governing body. Display a categorized list of articles/videos based on topics. Display charts or graphs dynamically and allow filters (time period, donor type, region). Query waitlist database for organ types and patient data (non-personal, location).	HF007 AD004
	PU002	Registration	Visitors can register to the platform under the relevant roll.	<u>Business Rules:</u> User provided email address must be unique and verified Password must be strong and secure (min 8 chars, mix of letters, numbers, and symbols) <u>Business Logic:</u> On registration, the system checks if email already exists in the database. If new, send a verification email with a unique token/link.	AD001

				Email must follow a valid format (e.g., username@example.com). Users cannot proceed until email is verified. Validate password format on input Confirm password must match Hash password before saving to database	
	PU003	Donor registration	Visitors can register as donors.	<u>Business Rules:</u> Donors must submit a valid NIC number. Donors must agree to terms and conditions. <u>Business Logic:</u> The registration form includes fields for NIC number. The system validates NIC number format (e.g., national NIC rules). Donors must tick a checkbox to accept terms and conditions. Mark Donor as “Pending Verification” until admin approval. Consent and health information are saved for admin review.	AD001
	PU004	Submit Feedback	Submit a system feedback form without logging in.	<u>Business Rules:</u> Spam prevention mechanisms must be enforced (e.g., CAPTCHA). Users must agree to basic terms (e.g., no hate speech, no personal data). Feedback should be non-identifiable unless the user voluntarily shares their contact	

				<u>Business Logic:</u> Feedback form includes optional name/email, message box, and rating. System validates for empty fields and proper formatting (if contact info is entered). Store feedback in the backend tagged as “Anonymous” if no contact given. Send acknowledgment message or confirmation popup after submission..	
Admin Users	AU001	Hospital registration	Register hospital with license and verification documents.	<u>Business Rules:</u> The hospital's physical address and official name must be provided. Hospitals must submit a valid medical registration/license number. The license submitted must not be expired at the time of registration. Hospitals must upload a copy of their official registration/license certificate. <u>Business Logic:</u> License number is validated using format (e.g., alphanumeric with length check). Mark hospital as “Pending Verification” until admin approval.	AD001
Donor	DF001	Login	Registered donors can login to the system and view their dashboard.	<u>Business Rules:</u> Only registered and verified users are allowed to log in to the system. At the first login, users must select their role(s) in the system. The system provides three possible roles:	AD001

				○ <u>Organ Donor</u> ○ <u>Financial Donor</u> ○ <u>Non-Donor</u> Users are allowed to select multiple roles where applicable. Organ Donor and Non-Donor roles cannot be selected simultaneously, as they represent conflicting statuses. Financial Donor can be combined with either Organ Donor or Non-Donor. Once roles are selected during the first login, the system dashboard and accessible features will be customized based on the selected roles. Users are allowed to view and update their personal profile information after logging into the system. <u>Business Logic:</u> The system verifies the login credentials (user ID and password) against the stored user records in the database. If the credentials are valid and the user account is verified, the system grants access to the login session. When a user logs in for the first time, the system checks whether the user has assigned roles in the database. If no roles exist, the system redirects the user to the role selection interface. The system validates the selected roles according to the following rules: Organ Donor and Non-Donor cannot be selected together. Financial Donor can be	
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				selected with either Organ Donor or Non-Donor. After successful validation, the selected roles are stored in the user role table in the database. During subsequent logins, the system retrieves the stored roles and generates the appropriate dashboard and system features based on those roles. The system allows users to update their profile details, and the updated information is stored in the database after validation.	
	DF002	<i>Change Role</i>	Registered users are allowed to modify their roles within the system according to defined restrictions related to organ donation consent status.	<u>Business Rules</u> Users are allowed to add or modify roles in the system. A user who currently has the Organ Donor role may change their role to Non-Donor only if no active or pending organ donation consents exist. If the user has existing organ donation consents, they must withdraw all consents before changing the role to Non-Donor. The system must ensure that Organ Donor and Non-Donor roles cannot exist simultaneously for the same user. When a user withdraws all organ donation consents, the Organ Donor role becomes eligible for removal. <u>Business Logic</u> When a user attempts to change their role, the system checks the current roles assigned to the user. If the user attempts to change from Organ Donor to Non-Donor, the system verifies whether the user has active or pending organ donation consents.	

				If no consents exist, the system updates the user role to Non-Donor. If consents exist, the system prevents the role change and prompts the user to withdraw all consents first. When the user withdraws all consents, the system updates the consent status in the database accordingly. After all consents are withdrawn, the system allows the role change from Organ Donor to Non-Donor. The system updates the user roles table and refreshes the user dashboard based on the updated role configuration.	
	DF003	Fill Consent Form	donors must fill the consent forms for each organ donation.	<u>Business Rules:</u> Only users who have selected the Organ Donor role are allowed to access and submit donation consent forms. The system provides three types of consent forms: • Living Organ Donation Consent Form • After Death Organ Donation Consent Form • Full Body Donation Consent Form Donors may submit one or more consent forms depending on their donation preference. For Living Organ Donation, the donor must: • Select the hospital requesting the organ. • Provide the names and NIC numbers of two witnesses. For After Death Organ Donation, the donor	DF001

				must: • Provide the names and NIC numbers of two witnesses. • Provide the names and NIC numbers of two custodians. For Full Body Donation, the donor must: • Select the medical school. • Provide the names and NIC numbers of two witnesses. • Provide the names and NIC numbers of two custodians. The consent form must contain the signatures of the donor and the required witnesses and custodians. The donor must download the generated form, obtain the required signatures, and upload the signed document to the system. Uploaded consent forms must be reviewed and approved by the Donation Administrator. A consent becomes active only after administrator approval. <u>Business Logic:</u> After login, the system checks the roles assigned to the user. If the user has the Organ Donor role, the system displays available donation options and consent forms. The system provides access to three consent forms: • Living organ donation • After death organ donation • Full body donation When Living Organ Donation is selected: • The system loads the living	
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				donation form. • The donor selects the hospital requesting the organ. • The donor enters the names and NIC numbers of two witnesses. When After Death Organ Donation is selected: • The system loads the after-death donation form. • The donor enters the names and NIC numbers of two witnesses. • The donor enters the names and NIC numbers of two custodians. When Full Body Donation is selected: • The system loads the full body donation form. • The donor selects the medical school. • The donor enters the names and NIC numbers of two witnesses. • The donor enters the names and NIC numbers of two custodians. The system automatically fills donor details such as donor ID using the logged-in user account. After the donor completes the form, the system generates a downloadable PDF consent form. The donor must download the form, obtain the required signatures, and upload the signed PDF. The system stores the uploaded document securely and marks the consent status as "Pending Approval." The Donation Administrator reviews the	
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				uploaded consent form. If the form is valid, the administrator approves the consent and updates the status to "Active." If the form is invalid or incomplete, the administrator rejects the consent and the donor is notified to resubmit the form.	
	DF004	<i>Download Donor Card</i>	Donors who have completed at least one valid donation consent are allowed to download their donor identification card from the system.	<u>Business Rules</u> Only users with the Organ Donor role are eligible to obtain a donor card. A donor must have at least one approved and active consent form to download a donor card. The donor card acts as an official identification indicating the individual's consent for donation. If no active consent exists, the donor cannot download the donor card. Each donor is allowed to download their donor card multiple times if required. <u>Business Logic</u> When a donor accesses the Donor Card section, the system checks the user's assigned roles. The system verifies whether the user has the Organ Donor role. The system then checks the database for approved and active consent records associated with the donor. If at least one active consent exists, the system enables the donor card generation option. The system generates the donor card using the donor's registered information (such as name, donor ID, and donation status).	

				The donor card is generated as a downloadable digital document (PDF). The donor can download and store the donor card for personal use or identification purposes.	
		<i>View Upcoming Appointments (Pre-Donation Tests)</i>	After submitting a Living Donation Consent Form, donors can view and manage their scheduled medical test appointments required for donor–recipient matching.	<u>Business Rules</u> Only users who have submitted a Living Organ Donation Consent Form are eligible to access appointments. Appointments are created by the hospital or medical authority for matching tests. Donors can view all upcoming appointments in a calendar view. Donors are allowed to accept or reject appointments. If an appointment is rejected, the donor must provide a valid reason. Rejected appointments must be rescheduled by the hospital. Donors cannot modify appointment details except accept/reject status. <u>Business Logic</u> After login, the system verifies whether the donor has a valid Living Donation Consent Form. If valid, the system retrieves all upcoming appointment records linked to the donor. The system displays appointments in a calendar-based interface. Donor selects an appointment to view details.	

				Donor can choose to: Accept appointment Reject appointment with reason If accepted, the system updates the appointment status to Confirmed. If rejected, the system: Stores rejection reason Updates status to Rejected Sends notification to hospital Hospital reviews rejection and reschedules appointment with a new date/time.	
		<i>View Test Results</i>	Donors can view their medical test results after hospitals upload and update them in the system. These tests are part of the donor–recipient matching and eligibility verification process.	<u>Business Rules</u> Only registered and verified donors can access test results. Donors can only view their own test results. Test results must be uploaded by authorized hospitals. Test results become visible only after they are successfully submitted and stored in the system. Donors cannot modify or delete test results. <u>Business Logic</u> The hospital uploads the donor’s medical test results into the system. The system validates and securely stores the uploaded results. When the donor logs in, the system	

				verifies their identity and role. The system retrieves all test results linked to the donor ID. The system displays available results to the donor. The donor can view or download the test results.	
	DF005	<i>View Approved Labs</i>	After approving the consent form and getting a request to donate, the donor should visit a lab and get relevant reports.	<u>Business Rules:</u> Only approved labs should be displayed. <u>Business Logic:</u> Allow filters by location or specialty.	DF001
	DF005	<i>Receive Organ Request Notification</i>	When a donor has an active organ donation consent, the system continuously checks for matching organ requests submitted by hospitals. If a match is found, the donor receives notifications of the requests and can choose to accept one of them.	<u>Business Rules</u> Only donors with an active organ donation consent are eligible to receive organ request notifications. The system may identify multiple matching organ requests from different hospitals. All matching requests are notified to the donor through the system. The donor is allowed to accept only one organ request. Once the donor accepts a request, all other pending requests for that organ automatically become rejected. If the donor rejects a request, the request status is updated as Rejected. Hospitals must be notified of the donor's response .	DF001 HF005

				<u>Business Logic</u> The system searches the organ request database for requests that match the donor's registered organ donation consent. If matching requests are found, the system generates notifications for the donor. The donor accesses the notification page and views all matching requests from hospitals. The donor can choose to accept or reject a request. If the donor accepts a request: The system updates the selected request status to Accepted. All other matching requests are automatically updated to Rejected. Notifications are sent to the respective hospitals. If the donor rejects a request: The system updates the request status to Rejected. Other pending requests remain available for the donor to respond.	
	DF006	<i>View Donation Status</i>	After donation done Its notify on the donor profile	<u>Business Rules:</u> Status should reflect real-time updates from the hospital. Donors can only view their own donation history. <u>Business Logic:</u> Display current stage: active, in-progress, completed, etc.	HF007

	DF007	<i>Receive Appreciation Certificate</i>	After donation is completed the donor will receive a certificate and rating for their profile.	<u>Business Rules:</u> Only issued if a donation has been completed. Certificates must include the donor name and date of donation. <u>Business Logic:</u> Make it downloadable as PDF.	DF001 DF005
	DF008	<i>After Care portal access</i>	After donation donors can access the aftercare portal for their annual checkups.	<u>Business Rules:</u> Available only for actual donors who donate through the system. The after care package depends on the donation type. <u>Business Logic:</u> Load form for after care. Match donor with correct care plan.	DF001 DF005
	DF009	<i>Update Donor Profile Details</i>	After verification donors can access their profile. (Donor dashboard)	<u>Business Rules:</u> Email and NIC cannot be changed after verification. <u>Business Logic:</u> Load editable fields (contact, address) and save changes.	DF001
	DF010	<i>Withdraw Donor Consents.</i>	After registration anytime donor can withdraw their registration	<u>Business Rules:</u> Donors can only withdraw if no active donation process is ongoing.	DF001

				Withdrawal must be confirmed by the admin. <u>Business Logic:</u> Check if any donation requests are pending. Update account status to "Withdrawn."	
	DF011	<i>Financial support Requests (donors who accept financial donor role)</i>	donors can see requests on their payment view.	<u>Business Rules:</u> donors can view all open support requests from patients/hospitals. Requests are categorized by support type. donors cannot see personal medical data. <u>Business Logic:</u> The system fetches all active requests tagged with categories. Display limited information: type of support needed, cost, urgency.	
	DF012	<i>Make Payment (donors who accept financial donor role)</i>	Financial supporters can donate their donations for the community.	<u>Business Rules:</u> F.donor must have valid and complete payment details. Payment should be linked to a specific support type or request. <u>Business Logic:</u> **_<dummy> System validates payment details and ensures all fields are complete.	
	DF013	<i>View financial</i>	Financial donors	<u>Business Rules:</u>	

		<i>donation history</i>	can view their donation history.	View history <u>Business Logic:</u> Fetch data from the database.	
		<i>Download financial donation Appreciation Certificate</i>	Donors who contribute financial donations to the system are eligible to receive appreciation certificates as recognition for their contributions.	<u>Business Rules</u> Only users who have the Financial Donor role are eligible to receive appreciation certificates. A financial donor must have completed at least one successful donation transaction to be eligible for a certificate. Donors are allowed to download an appreciation certificate for each individual donation. Donors are also allowed to download a certificate representing the total cumulative donation amount made to the system. Certificates can only be generated for confirmed and successfully recorded donations. Donors may download the certificates multiple times if required. <u>Business Logic</u> When a financial donor accesses the Appreciation Certificate section, the system verifies that the user has the Financial Donor role. The system retrieves the donor's donation records from the database. If at least one completed donation exists, the system enables the certificate generation options. The system provides two certificate options:	

				• Certificate for each individual donation • Certificate for the total cumulative donation amount When a donor selects the individual donation certificate, the system generates a certificate containing details such as: • Donor name • Donation amount • Donation date When a donor selects the total donation certificate, the system calculates the total amount donated by the donor and generates a certificate reflecting the cumulative contribution. The system generates the certificate as a downloadable PDF document. The donor can download and store the appreciation certificate for record or personal recognition.	
		Account Withdrawal (User Registration Termination)	Registered users are allowed to permanently withdraw their account from the system under controlled conditions	<u>Business Rules</u> Users are allowed to request account withdrawal at any time after registration. A user can only withdraw their account if there are no active or ongoing donation processes. Users with pending or active organ donation consents cannot withdraw their account unless all consents are withdrawn first. Account withdrawal must be confirmed and approved by the system administrator. Once an account is withdrawn, the user will lose access to all system features and	

				services. Withdrawal action is permanent and cannot be reversed without re-registration. <u>Business Logic</u> The system receives an account withdrawal request from the user. The system checks the user's current status and retrieves any active or pending donation records. If active processes exist, the system blocks the withdrawal request and notifies the user. If no active processes exist, the system forwards the withdrawal request to the administrator for approval. The administrator reviews the request and either approves or rejects it. If approved, the system updates the user status to "Withdrawn" in the database. The system disables user login and removes access to donor services. The user is notified of the successful account withdrawal.	
Hospital	HF001	<i>Login</i>	Registered hospitals can access the system and view the hospital dashboard.	<u>Business Rules:</u> Only registered and verified hospitals are allowed to log in. <u>Business Logic:</u> Login credentials (hospital license number and password) must match stored values.	AD001
	HF002	<i>Organ Request Management</i>	After registration, hospitals can	<u>Business Rules:</u> A hospital can add new organ requests,	HF001

			request organs through the system for their patients.	Update existing organ requests , Delete organ requests. Organ requests must include organ type and urgency. Only verified and logged-in hospitals can manage organ requests. <u>Business Logic:</u> Validate organ type and urgency from a predefined list.	
	HF003	<i>Donor Eligibility Update</i>	After medical tests the hospital can update donors eligibility.	<u>Business Rules:</u> Eligibility decisions include Approved, rejected. Updates must be documented with reasoning (medical condition, lab reports). <u>Business Logic:</u> Only registered and verified hospitals can access this data.	HF001
	HF004	<i>Donor Consent Management</i>	After donors submit the consent forms hospitals can see available organs.	<u>Business Rules:</u> Filtering helps in matching potential donors to recipients quickly. <u>Business Logic:</u> Hospitals can see only donor consent and can not see any donor details.(Organ type, blood group, age, location)	HF001 DF002
	HF005	<i>Send Transplant Requests to Donors. (After matching)</i>	After matching the donors' organs, the hospital can send a request for the donor to	<u>Business Rules:</u> The transplant request can only be initiated after a successful donor-recipient match.	HF001 HF004

			the transplant process.	<u>Business Logic:</u> Should depend on the organ matching system.	
	HF006	<i>Recipient-Patient registration</i>	After transplant operations hospitals can add recipient patients to the after care only if they need.	<u>Business Rules:</u> Only hospitals can register recipient patents. Hospitals must submit a valid patient registration number and patient id number. <u>Business Logic:</u> Verify the id number.	HF001
	HF007	<i>Update Donor Status</i>	After donation, hospitals can notify the system.	<u>Business Rules:</u> Hospitals can update the donation process outcome (Successful or Cancelled) <u>Business Logic:</u> Status can be marked as Successful only after a verified transplant.	HF001
	HF008	<i>Add Stories</i>	Hospitals can add stories and special notifications on the home page.	<u>Business Rules:</u> Only verified and logged-in hospitals can submit stories. All submitted stories must be reviewed and approved by an admin before publication. <u>Business Logic:</u> System displays form. Status should be "Pending Approval" until store is approved by the admin.	HF001

Financial donors	FD001	Login			AD001
	FDF002	Select Support Type			FD001
	FD003	Financial support Requests (donors who accept financial donor role)			FD001
	FD004	Make Payment			FD001
	FD005	View donation history	Financial donors can view their donation history.	<u>Business Rules:</u> View history <u>Business Logic:</u> Fetch data from the database.	FD001
Post-Donation Patient	PDF001	Login			DF001 HF007
	PDF002	Medical Checkup Scheduling			PDF001
	PDF003	Request Financial Support			POF001
	PDF004	View notification			
Recipient Patient	RF001	Login	Registered patience can login into their profile.	<u>Business Rules:</u> Login using id number and password. <u>Business Logic:</u> Match id number with the password.	HF006

	RF002	<i>Request Medicine Support.</i>	Recipient patients can request financial support for their medicine.	<u>Business Rules:</u> Only verified patents can be requested. Admin is responsible for reviewing and either approving or rejecting support. <u>Business Logic:</u>	RF001
	RF003	<i>View notification</i>	After approving the request, the patient can see a notification about their request.	<u>Business Rules:</u> All approved/rejected support requests must trigger a patient-facing notification. Notifications should be visible in the user dashboard and optionally sent via email. <u>Business Logic:</u> When an admin updates a request (Approve/Reject) builds a custom message Displays in Notifications panel (dashboard) and email inbox	RF001 AD008
User Management Admin	AD001	<i>User Management</i>	Manage user roles, status, and account access.	<u>Business Rules:</u> Only admins can view and manage all user roles (donor, patient, hospital, financial donor). Admin can edit, activate, or deactivate user accounts. <u>Business Logic:</u> On user registration, assign the role as "Pending Verification." Admin dashboard shows lists by role and status. Actions: Approve, Block/Deactivate.	

	AD002	User Verification & Approval	Verify user documents and approve or reject registration.	<u>Business Rules:</u> Each user type must upload required documents for verification. Admin must review and verify documents before granting access. Rejected verifications must include a reason. <u>Business Logic:</u> Display uploaded documents to admin with approve/reject buttons. On approval → update user status to “Verified.” On rejection → notify the user with reason.	
	AD003	Update Donor Eligibility & Status	Approve, reject, and update donor medical eligibility status.	<u>Business Rules:</u> Admin can approve or reject donor eligibility based on medical input. <u>Business Logic:</u> Display pending donor eligibility cases. Admin selects “Approve” or “Reject” System updates donor profile and triggers notification.	
	AD004	Story Management	When a hospital adds a store, before publishing, the admin should approve.	<u>Business Rules:</u> Only admins can approve or reject hospital-submitted stories. <u>Business Logic:</u> Stories must not contain confidential patient details unless consent is provided.	

	AD005	Admin handling the User Feedback	Review, manage, and respond to user-submitted feedback.	<u>Business Rules:</u> All users (donors, patients, hospitals) can submit feedback. Admin can view, filter, and respond to feedback. Serious complaints must be flagged and assigned for follow-up. Feedback cannot be deleted but can be marked as “Resolved” or “Pending”. <u>Business Logic:</u> System displays submitted feedback with filters (type, user role, date). Admin can mark feedback as reviewed, resolved, or forward it to another department. System stores admin actions (reviewed/resolved) in a log. Optional: send confirmation or follow-up response to the user.	
Aftercare management admin	AD004	Support Request Handling	Review, approve, or reject patient support fund requests.	<u>Business Rules:</u> One request per patient per support cycle is allowed. <u>Business Logic:</u> Fetch pending requests from post-donation and recipient patients. Admin can approve or reject.	
	AD005	Admin Notification	Send, manage,	<u>Business Rules:</u>	

		Handling (for Both Patient Types)	and track notifications for patient updates.	Notifications must be sent when a support request is approved, rejected, or requires revision. Notifications must only be visible to the relevant patient. Critical updates (e.g., checkup reminders, approvals) must require patient acknowledgment. Notifications must be stored and visible in the patient dashboard for at least 12 months. <u>Business Logic:</u> System listens for admin actions like “Approve”, “Reject”, or “Schedule Checkup”. Notification is shown in the patient’s dashboard under “Notifications”. If email alerts are enabled, the system also sends the message via email. Patients can open, read, and (if required) acknowledge the notification. System logs when the notification was read and acknowledged.	
	AD006				
Donation Admin	AD007	Donation management	After receiving the donor confirmation, the admin can contact the donor and relevant hospital.	<u>Business Rules:</u> Once a donor confirms, the system must automatically notify the admin details. Admin should be able to see the requested hospital. <u>Business Logic:</u>	DF004

				Admin must be authenticated and authorized to access relevant donor data. Admin must be authenticated and authorized to access hospital data.	
** Finance Admin	AD008	<i>Approve patient requests.</i>	Only financial supporters can approve the requests and send vouchers.	<u>Business Rules:</u> <u>Business Logic:</u>	